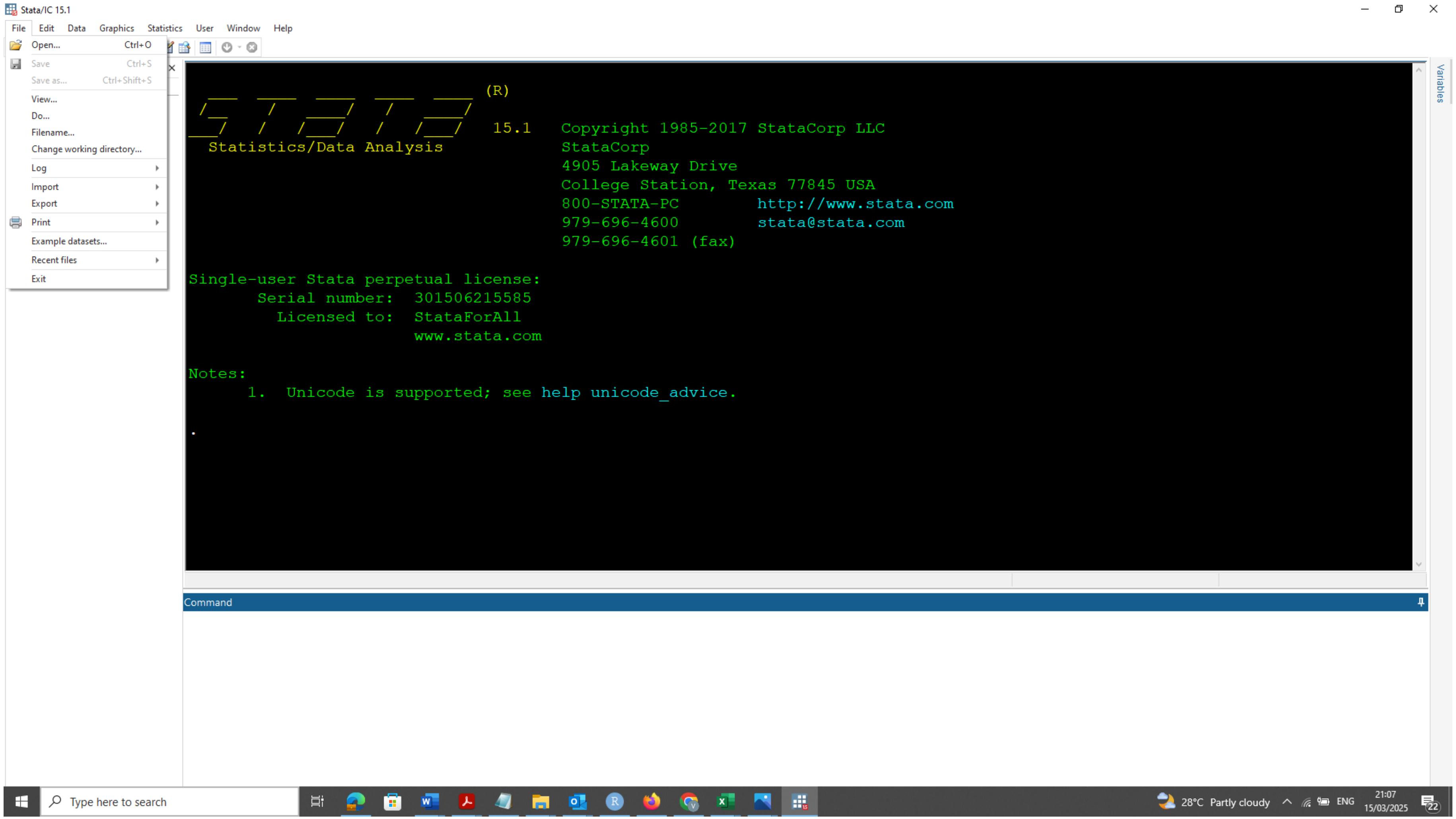
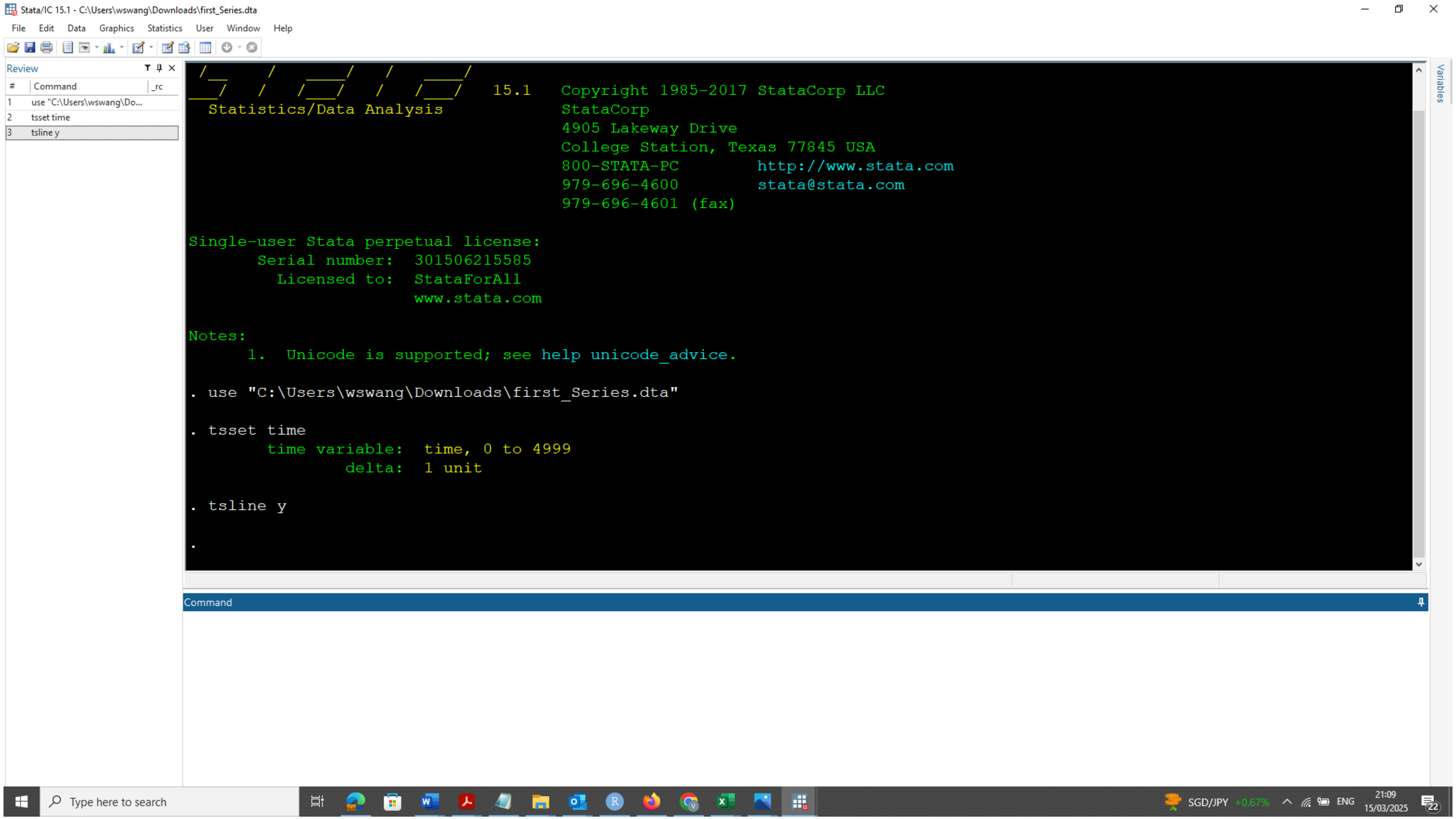
1

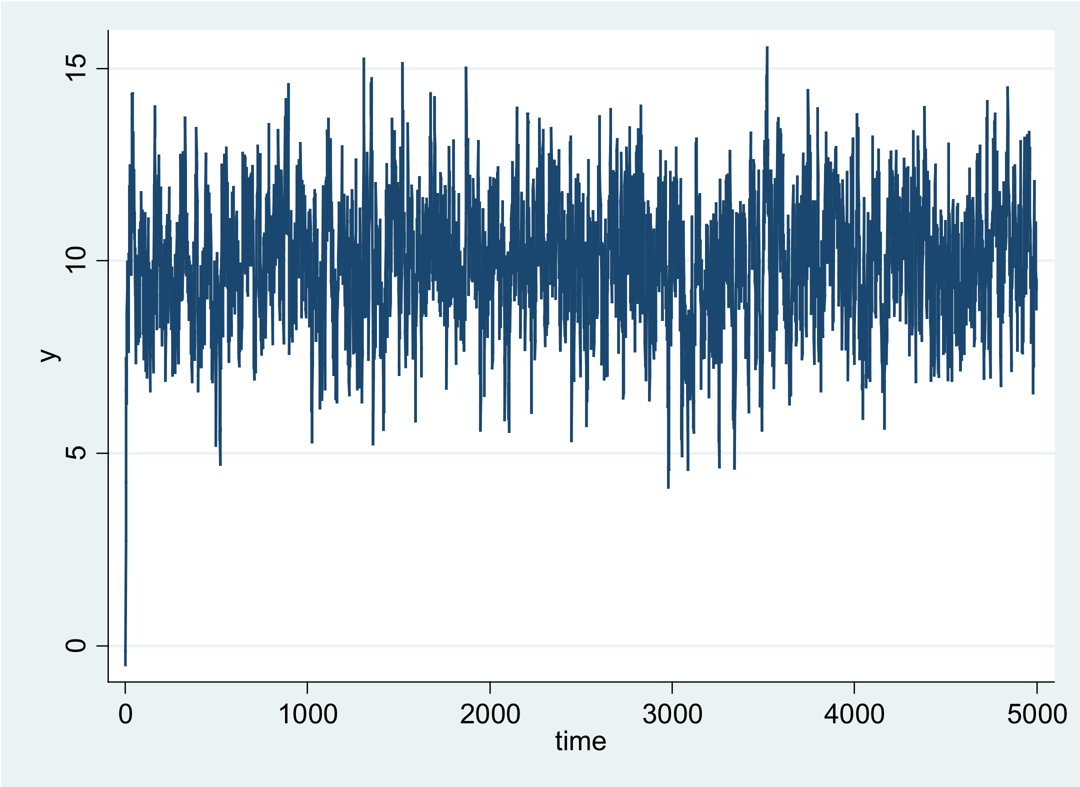
(a) We can open the data set (first series) through the open option.



Let Stata know that it is a time series dataset using 'tsset time'. Then use 'tsline' to create the time series line.



We can then quickly check the diagram to see whether the series is stationary. It appears that the series is more likely stationary because it has a constant mean (around 10) over time, and the variance is relatively stable over time, i.e., it fluctuates between 5 and 15 most of the time. However, to confirm its stationarity, we should use the unit root tests. Please note that for the stationary series, the covariance between y(t) and y(t-h) does not depend on t and cannot be checked from the diagram.



PP test

```
. pperron y

Phillips-Perron test for unit root                 Number of obs   =      4999
                                                   Newey-West lags =         9

                         ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(rho)        -1011.765          -20.700           -14.100           -11.300
 Z(t)            -24.042           -3.430            -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Reject the null hypothesis that the series is nonstationary

ADF test

```
. dfuller  y, lags(9)

Augmented Dickey-Fuller test for unit root         Number of obs   =      4990

                         ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -15.817           -3.430            -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

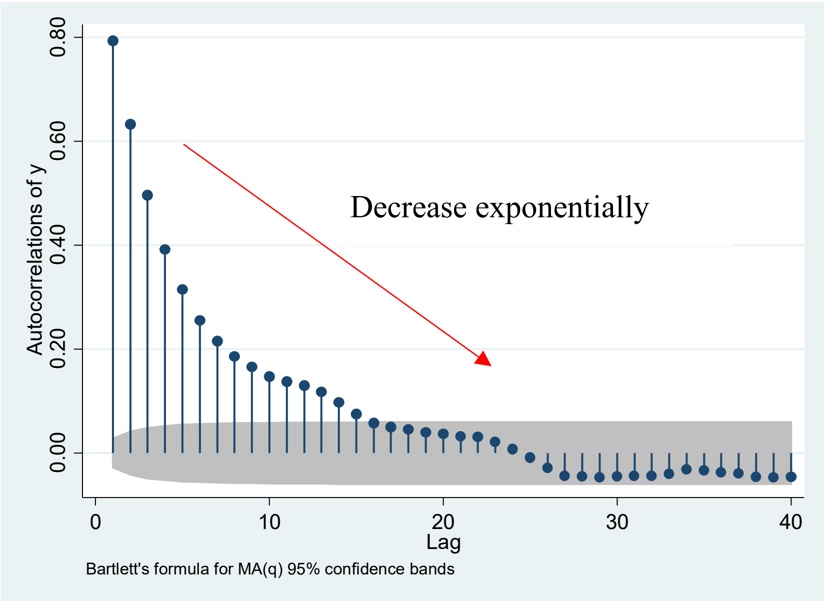
Note that, for the ADF test, I used lags (9) based on the suggestion from the PP test.

(b) Check autocorrelation function (ac). It appears that the bars drop to zero (i.e., the bars are in the grey area) quickly, unlike the McDonald's example discussed in class. If a series is nonstationary, the bars do not die down fast.

```
.
. tsset time
        time variable:  time, 0 to 4999
                delta:  1 unit

. tsline y

. ac y
```



(C)

```
. reg y l1.y

      Source |       SS       df       MS              Number of obs =    4999
-------------+------------------------------           F(  1,  4997) = 8706.31
       Model |  8372.15451     1  8372.15451           Prob > F      =  0.0000
    Residual |  4805.21145  4997  .961619262           R-squared     =  0.6353
-------------+------------------------------           Adj R-squared =  0.6353
       Total |   13177.366  4998   2.6365278           Root MSE      =  .98062

------------------------------------------------------------------------------
           y |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
           y |
         L1. |   .7938323   .0085077    93.31   0.000     .7771534    .8105111
             |
       _cons |   2.058418    .085992    23.94   0.000     1.889836       2.227
------------------------------------------------------------------------------
```

The estimated regression model is

$$\hat{y}_t = 2.05 + 0.793y_{t-1}$$

As we can see here, the estimated intercept is 2.05, which is very close to the true intercept value of 2, and the estimate slope 0.7938 is also very close to the true slope value of 0.8.

(d) Once we have done the estimation in (c), we can calculate the residuals and check if the residuals are white noise.

```
. predict uhat, r
(1 missing value generated)

. wntestq uhat

Portmanteau test for white noise
---------------------------------------
 Portmanteau (Q) statistic =    39.6316
 Prob > chi2(40)           =     0.4867
```

The null is that the residuals are white noise, and the alternative is the residuals are not white noise. Since the p value is 0.48 which is much larger than the 5% significance level, we cannot reject the null.

Remark:

As discussed during the tut, if the residuals are white noise, it implies Assumptions TS 4 and 5 hold.